

Chapter 7 Feynman Rules

Ch. 7 Feynman

- Structure of form

$$\langle f | S | i \rangle \sim \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle$$

w/ $|\Omega\rangle$ ground state

→ Feynman propagator
(Free theory & Arel/S)

$$D_F(x, y) \equiv \langle 0 | T \{ \phi(x_1) \phi(x_2) \} | 0 \rangle$$

$$= \lim_{\epsilon \rightarrow 0} \int \frac{d^4 k}{(2\pi)^3} \frac{1}{k^2 - m^2 + i\epsilon} e^{i k(x-y)}$$

w/ $|0\rangle$ free ground state

precisely:

- Time ordered products via integrals over Feynman propagators
- Perturbation w/ Feynman diagrams
- You specify Feynman Rules for Time ord. prod.
- momentum space for symmetry elements

1st:

- Feynman rules via Lagrangian & anotherology
- Quick way to connect Feynman diagrams to CT
- then Hamiltonian at same dep. prob. theory
- More exercises for QM
- connects time evolution to Hermitian Hamiltonian
- Path approximation again
- 2nd way is Path integral (disc 14 upcoming)

2.1 Lagrangian Derivation

Free quantum fields:

$$[\phi(\vec{z}, t), \phi(\vec{z}', t)] = 0$$

$$[\phi(\vec{z}, t), \partial_t \phi(\vec{z}', t)] = i\delta^3(\vec{z} - \vec{z}')$$

can be classical limit

free fields:

$$(\square + m^2)\phi = 0$$

"rel. to generators of symmetry transformations is determined?"

- can assume $i\partial_t \phi = [\phi, H]$

Lag. approach:

note: can only calculate

$\langle 0 | T \xi \phi(x) \phi(x') \xi | 0 \rangle$ in free theory is for

- useful for interacting:

$$\langle 0 | T \xi (\Box + m^2) \phi(x) \phi(x') \xi | 0 \rangle = \langle 0 | T \xi (\Box + m^2) \phi(x) \phi(x') | 0 \rangle - i \delta^4(x-x')$$

- $\delta^4(x-x')$ shows diff. between classical & quantum theories

permutation

$$\partial_t \langle 0 | T \xi \phi(x) \phi(x') | 0 \rangle = \partial_t [\langle 0 | \phi(x) \phi(x') | 0 \rangle \theta(t-t')]$$

$$+ \langle 0 | \phi(x) \phi(x') | 0 \rangle \theta(t-t')$$

$$= \langle 0 | T \xi \partial_t \phi(x) \phi(x') \xi | 0 \rangle + \langle 0 | \phi(x) \phi(x') | 0 \rangle \partial_t \theta(t-t') + \langle 0 | \phi(x) \phi(x') | 0 \rangle \partial_t \theta(t-t')$$

$$\left(\partial_t \theta(t-t') - \delta(t-t') \right) \langle 0 | T \xi \partial_t \phi(x) \phi(x') \xi | 0 \rangle + \delta(t-t') \langle 0 | [\phi(x), \phi(x')] | 0 \rangle$$

- last term vanishes since $\delta(t-t') \rightarrow t=t'$
 $\& [\phi(x), \phi(x')] = 0$
 at $t=t'$

∂_t^2 gives:

$$\partial_t^2 \langle \Omega | T \{ \phi(x) \phi(x') \} | \Omega \rangle$$

$$= \langle \Omega | T \{ \partial_t^2 \phi(x) \phi(x') \} | \Omega \rangle + \delta(t-t') \langle \Omega | [\partial_t \phi(x), \phi(x')] | \Omega \rangle$$

- 8 forces time equiv.

$$\rightarrow [\partial_t \phi(x), \phi(x')] = -i\hbar \delta^3(\vec{x} - \vec{x}')$$

$$\Rightarrow \partial_t^2 \langle \Omega | T \{ \phi(x) \phi(x') \} | \Omega \rangle = \langle \Omega | T \{ \partial_t^2 \phi(x) \phi(x') \} | \Omega \rangle - i\hbar \delta^4(\vec{x} - \vec{x}') \\ \text{Then eqn follows w/ } \partial_t^2$$

$\&$ implies

$$(\partial_z + m) \phi P_F(x, y) = -i\hbar \delta^4(x, y)$$

in free theory

notations:

$$\langle \dots \rangle = \langle \Omega | T \{ \dots \} | \Omega \rangle$$

$$\rightarrow (\partial_t + m^2) \langle \phi(x) \phi(x') \rangle = \langle (\partial_t + m^2) \phi(x) \phi(x') \rangle - i\hbar \delta^4(x-x')$$

more fields =

?

$$\square_n(\phi(x_1)\phi(x_2)\dots\phi(x_n)) = \langle \square_n \phi(x_1)\phi(x_2)\dots\phi(x_n) \rangle$$

$$= i^n \sum_j \delta^4(x-x_j) \langle \phi(x_1)\dots\phi(x_{j-1}) \times \phi(x_{j+1})\dots\phi(x_n) \rangle$$

(note to g.s.)

$$\text{calculate } \square_n \langle \phi(x_1)\phi(x_2)\phi(x_3) \rangle$$

Next: Quantum fields satisfy same E.o.m. as class. fields

$$\text{If } \mathcal{L} = -\frac{1}{2} (\partial_\mu \phi)^2 + \mathcal{L}_{int}[\phi] \text{ then } (\square + m^2)\phi - \mathcal{L}'_{int}[\phi] = 0$$

Then: Schrodinger - Dyson Equations

$$(\square + m^2) \langle \phi(x_1)\phi(x_2)\dots\phi(x_n) \rangle = \langle \mathcal{L}'_{int}[\phi] \phi_1 \dots \phi_n \rangle$$

$$= i^n \sum_j \delta^4(x-x_j) \langle \phi_1 \dots \phi_{j-1} \times \phi_{j+1} \dots \phi_n \rangle$$

$$w/ \phi_n = \phi(x_n)$$

$$\phi_j = \phi(x_j)$$

- Euclid's difference between classical & quantum

- in classical theory, $\hbar \rightarrow 0$ so $[d(\vec{r}, t), \phi(\vec{r}, t)] = 0$

so we get $\delta^4(x - x_j) = 0$

- In a classical theory, comm. functions
is actually some bottles as fields
within comm. functions

- For QFT we only up to δ func.
- called contact interactions

Contact interactions:

- allow virtual particles & closed loops

Efficiency/Notation:

$$\delta_{ii} = \delta^4(x - x_i), \quad D_{ij} = D_{ji} = D_F(x_i, x_j),$$

$m=0$ since $m \neq 0$ trivial to generalize, $\hbar \rightarrow 1$.

Ex: Feynman Propagator Green function:

$$\square_n D_m = -i \delta_{nm}$$

Ex²: 2 point function:

$$\begin{aligned}\langle \phi_1, \phi_2 \rangle &= \int d^4x \delta_{x1} \langle \phi_2 \phi_2 \rangle = i \int d^4x (D_{xx}) \langle \phi_2 \phi_2 \rangle \\ &= i \int d^4x D_{x1} D_{x2} \langle \phi_2 \phi_2 \rangle \\ &\leftarrow \text{integration by parts}\end{aligned}$$

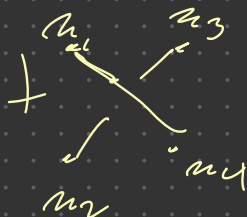
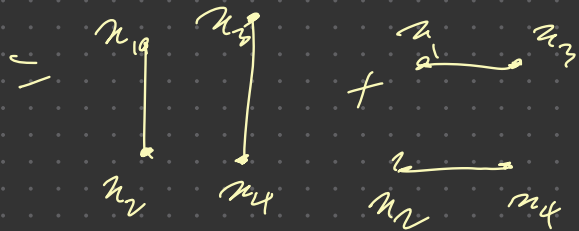
Free theory: (Lind = 0 3.D. $\Rightarrow$ $D_{xx} \langle \phi_2 \phi_2 \rangle = -i \delta_{xx}$)

$$\langle \phi_1, \phi_2 \rangle = \int d^4x D_{x1} \delta_{x2} = D_{12}$$

similar

$$\begin{aligned}\langle \phi_1, \phi_2, \phi_3, \phi_4 \rangle &= i \int d^4x D_{x1} D_{x2} \langle \phi_3 \phi_4 \phi_3 \phi_4 \rangle \\ &= \int d^4x \{ \delta_{x2} \langle \phi_3 \phi_4 \rangle + \delta_{x3} \langle \phi_2 \phi_4 \rangle \\ &\quad + \delta_{x4} \langle \phi_2 \phi_3 \rangle \} \end{aligned}$$

$$\langle \phi_1, \phi_2, \phi_3, \phi_4 \rangle = D_{12} D_{34} + D_{13} D_{24} + D_{14} D_{23}$$



Interaction terms:

2 point function w/

$$\mathcal{L} = -\frac{1}{2} \phi \square \phi + \frac{g}{3!} \phi^3$$

(w/ convention)

$$\mathcal{L}'_{int}[\phi] = \frac{g}{3!} \phi^3$$

So $\langle \phi_1 \phi_2 \rangle = i \int d^4 x \delta_{12} \langle \frac{g}{2} \phi^2 \phi_2 \rangle - i \delta_{12}$

???

Another integral ?

$$\rightarrow (\phi_1, \phi_2) = P_{12} - \frac{g}{2} \int d^4x d^4y P_{11} P_{22} \phi_y$$

$$\times (\phi_x^2 \phi_y)$$

~~???~~

$$= P_{12} - \frac{g^2}{4} \int d^4x d^4y P_{11} P_{22} (\phi_x^2 \phi_y)$$

$$+ y \int d^4x P_{11} P_{22} (\phi_x)$$

π $\rightarrow$ S.D. $\rightarrow$

$$(\phi_x^2 \phi_y^2) = 2P_{xy}^2 + P_{xx} P_{yy} + O(g)$$

$$(\phi_2) = i \int d^4y P_{yy} \square_y (\phi_y)$$

$$= i \frac{g}{2} \int d^4y P_{yy} (\phi_y^2)$$

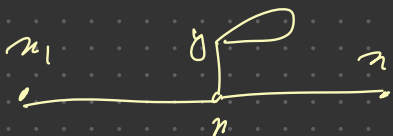
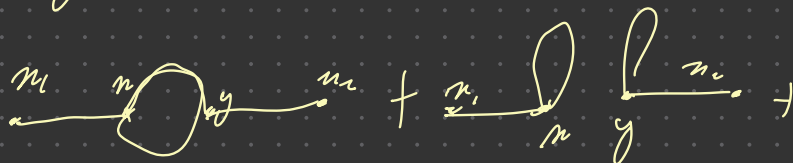
$$= i \frac{g}{2} \int d^4y P_{xx} P_{yy} + O(g^2)$$

$$(\phi_1, \phi_2) = P_{12} - g^2 \int d^4x d^4y \left(\frac{1}{2} P_{11} P_{22} \phi_y^2 \right.$$

$$+ \frac{1}{4} P_{11} P_{xx} P_{yy} P_{yy}$$

$$+ \frac{1}{2} P_{11} P_{22} P_{xy} P_{xy} \Big)$$

Diagrams:



Feynman Rules:

1) start with external points

n_i & draw lines backward

2) line can either contract into existing line or Feynman propagator

- or can split b.c. of interaction

- split gives new internal vertex prop. to coeff. of $L'_{int}[\phi]$ & new lines corr. to fields

$L'_{int}[\phi]$

3. at given order, result is sum of all lines contracted & integrated over internal vertices.

- position space Feynman rules

- Time prod. det. by sum over diagrams / integrals up to numerical factor

- To determine convention to write by # permutations identical fields

Ex:

$$\text{Limb} \supset \frac{\lambda}{4!} \phi^4, \frac{g}{3!} \phi^3, \frac{\kappa}{5! 2! 2!} \phi^5 \phi_2^2 \phi_3^2 \dots$$

- when derivative taken, flux lines factor $\frac{1}{(n-1)!}$ cancelled by # permutations at lines out of vertex

→ $n!$ factors cancel & so already have prefactors λ, g, κ , etc)

- some cases permutations have same amplitude so must bin by 2 to get prefactor

Rule 4) Prop all $n!$ factors in coeff. flow th. by geometric sym factor

- Symmetry factor =

of ways graph can be deformed w/ n lines

- Compare & contrast of classical & quantum

Classical: no contact interactions

- Allowed to contract 2 fields to use 1-particle fields
(e.g. $\square \langle \phi_1 \phi_2 \phi_3 \phi_4 \rangle = i \delta_{12} \langle \phi_3 \phi_4 \rangle + \dots$)

Classical

→ only 'grad. func':
convergence?

(allowed)

Problems:



(the loops or virtual particle creation & destruction)

2.2 Hamiltonian Persp.

Heisenberg EOM:

$$i\partial_t \phi(x) = [\phi, H]$$

$$\rightarrow \phi(\vec{x}, t) = S(t, t_0)^\dagger \phi(\vec{x}) S(t, t_0)$$

w/ 3-momenta

acting

$$i\partial_t S(t, t_0) = H(t) S(t, t_0)$$

- Heisenberg picture

- states w/ $|\Omega\rangle$ time indep.

don't need explicit $H(t)$, just time dep. $H(t)$

write $H(t) = H_0 + V(t)$

ex: ϕ^3 interaction:

$$V(t) = \int d^3x \, g/2! \, \phi(\vec{x}, t)^3$$

interaction picture:

(fields only evolve w/ H_0)

free fields:

$$\phi_0(\vec{x}, t) = e^{iH_0(t-t_0)} \phi(\vec{x}) e^{-iH_0(t-t_0)}$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p e^{-ipx} + a_p^\dagger e^{ipx} \right)$$

($\phi(\vec{x})$ qchröd. free field)

$$\rightarrow \phi(\vec{r}, t) = U^\dagger(t, t_0) e^{-iH(t, t_0)} \phi_0(\vec{r}, t) e^{iH_0(t-t_0)} S(t, t_0)$$

$$= U^\dagger(t, t_0) \phi_0(\vec{r}, t) U(t, t_0)$$

w/ $U(t, t_0) \equiv e^{iH_0(t-t_0)} S(t, t_0)$ = unitary time evolution

- relates Heisenberg picture free fields at same time

$$i\partial_t U(t, t_0) = i(\partial_t e^{iH_0(t-t_0)}) S(t, t_0) + e^{iH_0(t-t_0)} i\partial_t S(t, t_0)$$

$$= - e^{iH_0(t-t_0)} H_0 S(t, t_0)$$

$$+ e^{iH_0(t-t_0)} H(t) S(t, t_0)$$

$$= e^{iH_0(t-t_0)} [-H_0 + H(t)] e^{iH_0(t-t_0)} S(t, t_0)$$

$$= U_I(t) U(t, t_0)$$

$$U_I(t) \equiv e^{iH_0(t-t_0)} U(t) e^{-iH_0(t-t_0)} \quad \text{is}$$

Heisenberg potential in interaction picture

bad if commuted:

$$U(t, t_0) = \exp\left(-i \int_{t_0}^t V_I(t') dt'\right)$$

- but the case since $[V_I(t_1), V_I(t_2)] \neq 0$

- similar for U

$$U(t, t_0) = T \left\{ \exp \left[-i \int_{t_0}^t dt' V_I(t') \right] \right\}$$

w/ T time order since it involves $\hbar$ & commutes

- note: $U(t, t) = 1$ so correct

Expansion: (since exp)

$$U(t, t_0) = 1 - i \int_{t_0}^t dt' V_I(t') - \frac{1}{2} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' T \{ V_I(t') V_I(t'') \} + \dots$$

Dyson Series

Perturbation for Dyson Series:

w/ $V = V_I$ for efficiency

$$i \partial_t U(t, t_0) = V(t) U(t, t_0)$$

$$\rightarrow U(t, t_0) = 1 - i \int_{t_0}^t dt' U(t') V(t', t_0)$$

imp (being correct) s. t_0 $U(t, t) = 1$

1st order approx:

$$U(t, t_0) \approx 1$$

1st order in V :

$$U(t, t_0) = 1 - i \int_{t_0}^t dt' U(t') V(t', t_0) + \dots$$

2nd order in V :

$$U(t, t_0) = 1 - i \int_{t_0}^t dt' U(t') V(t', t_0) \left[1 - i \int_{t_0}^{t'} dt'' U(t'') V(t'', t_0) + \dots \right]$$

$$= \left(1 - i \int_{t_0}^t dt' U(t') V(t', t_0) \right) + (-i)^2 \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' U(t') V(t', t_0) U(t'') V(t'', t_0) + \dots$$

Def: (chronology):

$t_0 < t'' < t' < t$ equivalent $t_0 < t'' < t$ and

$t'' < t' < t$ so

$$\int_{t_0}^t dt' \int_{t_0}^{t'} dt'' U(t') V(t'', t_0) U(t'') = \int_{t_0}^t dt'' \int_{t''}^t dt' U(t') V(t'', t_0) U(t'')$$

$$= \int_{t_1}^{t_2} dt'' \int_{t_0}^{t_1} dt' L(t'') V(t')$$

where $t'' \rightarrow t'$ and swapped order to get next term

Averaging gives:

$$\begin{aligned} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' L(t') V(t'') &= \frac{1}{2} \int_{t_0}^t dt' \left[\int_{t_0}^{t'} dt'' L(t') V(t'') + \int_{t_0}^{t'} dt'' L(t'') V(t') \right] \\ &= \frac{1}{2} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' T \{ L(t') V(t'') \} \end{aligned}$$

Then

$$V(t, t_0) = (-i) \int_{t_0}^t dt' L(t') + \frac{(-i)^2}{2} \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' T \{ L(t') L(t'') \} + \dots$$

$$\hookrightarrow V(t, t_0) = T \exp \left[-i \int_{t_0}^t dt' L_I(t') \right]$$

Verification:

$$V_{21} = V(t_2, t_1) = T \left\{ \exp \left[-i \int_{t_1}^{t_2} dt' \frac{1}{2} L(t') \right] \right\}$$

- "order doesn't matter" $\hookrightarrow$ $2L(t')$

$$\rightarrow V_{21} V_{12} = 1$$

$$L_{21}^{-1} = L_{21}^\dagger = V_{12}$$

w/ $\phi_1, L_{21} < L_3$:

$$L_{32} U_{21} = U_{31} \\ (x U_{12} \quad x U_{12})$$

$$\Rightarrow U_{31} U_{12} = U_{32}$$

$$\text{since } \phi(\vec{a}, t) = U^\dagger(t, t_0) \phi_0(\vec{a}, t_0) U(t, t_0)$$

$$\Rightarrow \phi(a_1) = \phi(\vec{a}, t_1) = U_{10}^\dagger \phi_0(\vec{a}, t_1) U_{10} = U_{c1} \phi_0(a_1) U_{10}$$

7.23 Vacuum Matrix Elements

- $U_{21} |\Omega\rangle$ was annihilated by $a_p(t)$ (at $t \rightarrow \infty$)

want to relate to ϕ that we know how free annihil & create operators move

$\rightarrow$ evolve to t_0 where interacting & free equal
Schrödinger

states evolve w/ $\phi(t, t_0) \rightarrow \phi(t, t_0) |\Omega\rangle$ annihil. by $a_p(t)$ at $t \rightarrow \infty$
reasoning:

$a_p(t)$ evolves like $[a_p, H]^\dagger a_p(t_0) \phi(t, t_0)$ & annihilates

$|\Omega\rangle$ at $t \rightarrow \infty$

Free: $|0\rangle$ annihil. by a_p

- a_p evolve w/ phase w/.

$\rightarrow a_p$ still annihil. $|0\rangle$ at any time

- Even if $|0\rangle$ doesn't have a energy,

$$a_p(t_0) e^{iH_0(t-t_0)} |0\rangle = 0 \text{ at } t \rightarrow \infty$$

since at $t=t_0$, the wavefunction

is arbitrary

$$e^{iH_0(t-t_0)} | \psi \rangle = \hat{U}_0(t, t_0) | \psi \rangle$$

$\rightarrow$ must be proportional states

$$\langle \Omega | \psi \rangle = N_i \lim_{t \rightarrow -\infty} \langle \psi(t, t_0) | e^{iH_0(t-t_0)} | \psi \rangle = N_i \langle \psi_0 | \psi \rangle$$

for some number N_i

g.i.w.

$$\langle \Omega | = N_f \langle \psi_0 | \text{ for some } N_f$$

Time ordered prod:

wlog $t_1 > \dots > t_n$

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \langle \Omega | \phi(x_1) \dots \phi(x_n) | \Omega \rangle$$

$$= N_i N_f \langle \psi_0 | U_{\infty,0} U_{0,t_1} \phi(x_1) U_{t_1,0} U_{0,t_2} \phi(x_2) U_{t_2,0} \dots U_{0,t_n} \phi(x_n) U_{t_n,0} U_{0,-\infty} | \psi_0 \rangle$$

$$= N_i N_f \langle \psi_0 | U_{\infty,0} \phi(x_1) U_{12} \phi(x_2) U_{23} \dots U_{(n-1)n} \phi(x_n) U_{n,-\infty} | \psi_0 \rangle$$

- Every thing in time order

therefore,

$$\langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle$$

$$= N_i N_f \langle \psi_0 | T \{ U_{\infty,0} \phi(x_1) U_{12} \phi(x_2) U_{23} \dots \phi(x_n) U_{n,-\infty} \} | \psi_0 \rangle$$

$$= N_i N_f \langle \psi_0 | T \{ \phi(x_1) \dots \phi(x_n) U_{\infty,-\infty} \} | \psi_0 \rangle$$

$$\text{since } \langle \Omega | \Omega \rangle = 1, N_i N_f = \langle \psi_0 | U_{\infty,-\infty} | \psi_0 \rangle$$

$$\rightarrow \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle = \frac{\langle \psi_0 | T \{ \phi(x_1) \dots \phi(x_n) U_{\infty,-\infty} \} | \psi_0 \rangle}{\langle \psi_0 | U_{\infty,-\infty} | \psi_0 \rangle}$$

$$\rightarrow \langle 0 | T \{ \phi(x_1) \dots \phi(x_n) \} | 0 \rangle =$$

$$\frac{\langle 0 | T \{ \phi_0(x_1) \dots \phi_0(x_n) \} \exp \left[-i \int_{-\infty}^{\infty} dt L_I(t) \right] | 0 \rangle}{\langle 0 | T \{ \exp \left[-i \int_{-\infty}^{\infty} dt L_I(t) \right] \} | 0 \rangle}$$

Interaction potential:

(cubic)

$$V(\phi) = \int d^3x \frac{g}{3!} \phi(\vec{x}, t_0) = \int d^3x \frac{g}{3!} \phi_0(\vec{x}, t) \\ = \int d^3x \frac{g}{3!} \phi(\vec{x})^3$$

Since:

$$\phi_0(\vec{x}, t) = e^{iH_0(t-t_0)} \phi_0(\vec{x}) e^{-iH_0(t-t_0)}$$

$$\therefore V_I = e^{iH_0(t-t_0)} \left[\int d^3x \frac{g}{3!} \phi_0(\vec{x})^3 \right] e^{-iH_0(t-t_0)} \\ = \int d^3x \frac{g}{3!} \phi_0(\vec{x}, t)^3$$

- Expressed in terms of free fields at all times

Back to Lagrangian

$$V_I = - \int d^3x \mathcal{L}_{int}[\phi_0] \\ \rightarrow V_{\infty, -\infty} = \exp \left[-i \int_{-\infty}^{\infty} dt L_I(t) \right] = \exp \left[i \int_{-\infty}^{\infty} dt \mathcal{L}_{int}[\phi_0] \right]$$

Summary:

$$\langle \Omega | \phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\langle 0 | U_{\infty} \phi_0(x_1) U_{1,2} \phi_0(x_2) U_{2,3} \dots \phi_0(x_n) U_{n,\infty} | 0 \rangle}{\langle 0 | U_{\infty, -\infty} | 0 \rangle}$$

w/ $|\Omega\rangle$ ground,

$$L_{ij} = T \left\{ \exp \left[i \int_{t_i}^{t_j} d^4x \mathcal{L}_{int}[\phi_0] \right] \right\}$$

$$(L_{int} = \mathcal{L} - \mathcal{L}_0 \text{ w/ } \mathcal{L}_0 \text{ free lag})$$

Time ordered operators:

$$\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\langle 0 | T \phi_0(x_1) \dots \phi_0(x_n) e^{i \int d^4x \mathcal{L}_{int}[\phi_0]} | 0 \rangle}{\langle 0 | T e^{i \int d^4x \mathcal{L}_{int}[\phi_0]} | 0 \rangle}$$

very nice

ϕ^3 theory examples

$$L_{int}[\phi] = \frac{\partial}{\partial \phi} \phi^3$$

$$\text{want } \langle \Omega | T \phi(x_1) \phi(x_2) | \Omega \rangle$$

perturbation in g .

$$\langle 0 | T \phi_0(x_1) \phi_0(x_2) e^{i \int d^4x \mathcal{L}_{int}[\phi_0]} | 0 \rangle$$

$$= \langle 0 | T \phi_0(x_1) \phi_0(x_2) | 0 \rangle$$

$$+ i \frac{\partial}{\partial \phi_1} \int d^4x \langle 0 | T \phi_0(x) \phi_0(x_1) \phi_0(x_2) | 0 \rangle$$

$$\left(\frac{ig}{32} \right)^2 \frac{1}{2} \int d^4x \int d^4y \quad 20 \left(T \frac{1}{2} \phi_0(x_1) \phi_0(x_2) \right. \\ \left. \times \phi_0(x)^3 \phi_0(y)^3 \right) |0\rangle$$

+ ...

- need to eval. correlation functions of prod. of free fields

$$\phi_0(x) = \phi_+(x) + \phi_-(x)$$

$$m/\phi_+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} a^\dagger_p e^{ipx} \quad \phi_-(x) = \int \frac{d^3p}{(2\pi)^3} \\ \times \frac{1}{\sqrt{2\omega_p}} a_p e^{-ipx}$$

ϕ_+ only creation, ϕ_- only annihilation

product ϕ_0 becomes sum of prod of ϕ_+ & ϕ_-

Ex:

$$20 \left| T \frac{1}{2} \phi_0(x_1) \phi_0(x_2) \phi_0(x)^3 \phi_0(y)^3 \right| 0 \rangle$$

$$= 20 \left| T \left\{ [\phi_+(x_1) + \phi_-(x_1)] [\phi_+(x_2) + \phi_-(x_2)] \right. \right. \\ \left. \left. \times [\phi_+(x) + \phi_-(x)]^3 [\phi_+(y) + \phi_-(y)]^3 \right\} \right| 0 \rangle$$

$$= 20 \left| T \frac{1}{2} \phi_+(x_1) \phi_+(x_2) \phi_+(x)^3 \phi_+(y)^3 \right| 0 \rangle$$

$$+ 20 \left| T \frac{1}{2} \phi_+(x_1) \phi_+(x_2) \phi_+(x) \phi_+(x)^3 \phi_+(y)^2 \phi_-(y) \right| 0 \rangle$$

+ ...

$-\phi_+$ creates particle & annihilated by ϕ_- ?

- pair of ϕ_+ w/ ϕ_- gives Feynman prop & is called contraction

contraction = creation then annihilation of particle,
gives Feynman Prop:

$$\langle 0 | T \phi_+(x) \phi_-(y) | 0 \rangle = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{ik(x-y)}$$

$$= D_F(x, y)$$

time ordered cov. we mean by sum over all possible
ways in which all of the fields in the product
can be contracted w/ each other!
(wicks thm in appdx)

$$D_{ij} = D_F(x_i, x_j)$$

Application of Wick's:

$$\text{first term of } \langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle$$

$$\text{is } \langle 0 | \phi_+(x_1) \phi_-(x_2) | 0 \rangle \text{ gives } D_{12}$$

2nd term odd fields, must disappear
 2nd term of fields

$$\rightarrow \langle 0 | T \sum \phi_0(x_1) \phi_0(x_2) \phi_0(x) \phi_0(x) \phi_0(y) \phi_0(y) \phi_0(y) \phi_0(y) \rangle \quad (10)$$

$$= 9 D_{12} D_{22} D_{2y} D_{yy} + 6 D_{12} D_{2y}^3$$

$$+ 18 D_{12} D_{22} D_{2y} D_{yy} + 9 D_{22} D_{2y} D_{22} D_{yy} +$$

$$18 D_{12} D_{2y} D_{2y}^2 + 18 D_{1y} D_{2y} D_{yy} D_{2y} D_{22}$$

$$+ 9 D_{1y} D_{22} D_{22} D_{yy} + 18 D_{1y} D_{22} D_{2y}^2$$

have to integrate to many Green's functions.

$$\langle 2 | T \sum \phi(x_1) \phi(x_2) \phi(x) \phi(y) \rangle = \frac{1}{\langle 0 | T \sum \phi(x_1) \phi(x_2) \phi(x) \phi(y) \rangle} \{ D_{12}$$

$$+ \int d^4x \left[\frac{1}{8} D_{12} D_{22} D_{2y} D_{yy} + \frac{1}{12} D_{12} D_{2y}^3 \right.$$

$$+ \frac{1}{2} D_{12} D_{22} D_{2y} D_{yy}$$

$$\left. + \frac{1}{12} D_{12} D_{22} D_{2y}^2 D_{yy} + \frac{1}{2} D_{12} D_{2y}^3 D_{yy} \right\}$$

- sum of terms is same as Feynman diagram approach
except
 $\langle 0 | T \xi e^{i \int d^4x \mathcal{L}_I(x)} | 0 \rangle$ and $\frac{1}{8} p_{12} p_{34} p_{56} + \frac{1}{12} p_{12}^3 p_{34}$

new diagram comes in, "bubbles" don't disappear



Cancellation

$$\langle 0 | T \xi e^{i \int d^4x \mathcal{L}_I(x)} | 0 \rangle = \langle 0 | 0 \rangle + \left(\frac{i g}{3!} \right)^2 \frac{1}{2} \int d^4x \int d^4y$$

$$\times \langle 0 | T \xi^2 \phi(x) \phi(y)^3 | 0 \rangle + \dots$$

(0 loop) term cancels by
miraculously

then

$$\hookrightarrow \langle 0 | T \xi e^{i \int d^4x \mathcal{L}_I(x)} | 0 \rangle = 1 + \left(\frac{i g}{3!} \right)^2 \frac{1}{2}$$

$$\times \int d^4x \int d^4y [9 p_{12} p_{34} p_{56} + 6 p_{12}^3]$$

- These are the

ϕ & θ bubbles

$$+ O(g^3)$$

we then find:

$$\frac{\langle 0 | T \sum \phi_a(x_1) \phi_a(x_2) e^{i \int \mathcal{L}_{int} \phi} | 0 \rangle}{\langle 0 | T \sum e^{i \int \mathcal{L}_{int} \phi} | 0 \rangle}$$

$$= \frac{D_{12} - g^2 \int \left[\frac{1}{8} D_{12} D_{xy} D_{xy} + \frac{1}{12} D_{12} D_{xy}^3 + \dots \right]}{1 - g^2 \int \left[\frac{1}{8} D_{12} D_{xy} D_{xy} + \frac{1}{12} D_{12} D_{xy}^3 \right]}$$

i.e. $\frac{1}{1+g^2 n} \approx 1 - g^2 + O(g^4)$, denominator cancels w/
numerator bubbles

- sum of numerator then is sum w/o bubbles $\times$ sum of bubbles
i.e.

$$\text{---} + \text{---} \bigcirc \text{---} + \frac{\text{---} \bigcirc \text{---}}{\text{---}} + \frac{\text{---} \bigcirc \text{---}}{\text{---}} + \dots$$

$$= (\text{---} + \text{---} \bigcirc \text{---} + \dots) \times (\text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} + \dots)$$

sum over bubbles is $\langle 0 | T \sum e^{i \int \mathcal{L}_{int} \phi} | 0 \rangle$

$$\text{A } 2\Omega | T \sum \phi(x_1) \dots \phi(x_n) \phi(x) | \Omega \rangle = 2\Omega | T \sum \phi(x) \phi(x) | \Omega \rangle$$

(no bubbles $\rightarrow$ every connected subgraph involves an external point) $\frac{\langle e^{i \int \mathcal{L}_{int} \phi} \rangle}{\langle e^{i \int \mathcal{L}_{int} \phi} \rangle}$ no bubbles

for 2-particle external states:

(with 4 external lines):

$$-\frac{1}{m_1} \text{ from external } = \left\{ \frac{1}{m_1} (iL_{m_1}) \right\}$$

cancel w/ m_1 from diagram prescription

- $\frac{1}{j!}$ for each interaction w/ j identical particles.

cancelled by $j!$ ways permuting j identical lines coming out of internal vertex

1.3 momentum space:

Diagram:

$$T_1 = \text{diagram} = -\frac{g^2}{2} \int d^4 x \int d^4 y D_{12} D_{34} \delta_{xy}$$

to eval: $D_{xy} = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 + i\epsilon} e^{ip(x-y)}$

$$\rightarrow T_1 = -\frac{g^2}{2} \int d^4 x \int d^4 y \left(\int \frac{d^4 p_1}{(2\pi)^4} \int \frac{d^4 p_2}{(2\pi)^4} \right) \frac{d^4 p_3}{(2\pi)^4}$$

$$\times \int \frac{d^4 p}{(2\pi)^4}$$

$$\times e^{ip_1(x-m)} e^{ip_2(y-m)} e^{ip_3(x-y)} e^{ip_4(x-y)} \frac{i}{p_1^2 + i\epsilon} \frac{i}{p_2^2 + i\epsilon} \frac{i}{p_3^2 + i\epsilon} \frac{i}{p_4^2 + i\epsilon}$$

in the integrals produce $\delta^4(-p_1 - p_2 + p_3)$ & $\delta^4(p_2 - p_3 - p_4)$

→ replace $p_3 \rightarrow p_1 - p_4$ & $\delta^4(p_2 - p_3 - p_4)$
becomes $\delta^4(p_1 - p_2)$

then set $p_4 = p$

$$\gamma_i = -\frac{\lambda^2}{2} \int \frac{d^4 k}{(2\pi)^4} \int \frac{d^4 p_1}{(2\pi)^4} \int \frac{d^4 p_2}{(2\pi)^4} e^{ip_1 x_1} e^{-ip_2 x_2} \\ \times \frac{i}{p_1^2 + i\epsilon} \frac{i}{p_2^2 + i\epsilon} \frac{i}{(p_1 - p)^2 + i\epsilon} \frac{i}{k^2 + i\epsilon} (2\pi)^4 \delta^4(p_1 - p_2)$$

Then let $q = z$:

$$\langle f | S | i \rangle = \left[-i \int d^4 x_1 e^{-ip_1 x_1} (p_1^2) \right] \left[-i \int d^4 x_2 e^{ip_2 x_2} (p_2^2) \right]$$

$$\times \langle \Omega | T \{ \phi(x_1) \phi(x_2) \} | \Omega \rangle$$

w/ p_1^2 & p_2^2 i.u. of the momenta

That's why we get:

$$2f | S | i \rangle = - \int d^4 x_1 e^{-ip_1 x_1} (p_1^2) \int d^4 x_2 e^{ip_2 x_2} (p_2^2) T_{12}$$

$$n.e. \text{ gives } (2\pi)^4 \delta(p_1 - p_1) \text{ & } n.e. \text{ gives } (2\pi)^4 \delta(p_2 - p_2) \neq 0$$

$$\langle f | S | i \rangle = -\frac{i\lambda^2}{2} \int \frac{d^4 k}{(2\pi)^4} \frac{i}{(p_i - k)^2 + i\epsilon} \frac{i}{k^2 + i\epsilon} (2\pi)^4 \delta^4(p_i - p_f) + \dots$$

- typically this integral is infinite so have to regularize

- $\delta^4(p_i - p_f)$ forces p conservation

note:

$$S = 1 + (2\pi)^4 \delta^4(p_i - p_f) i\mathcal{M}$$

$$\rightarrow i\mathcal{M} = -\frac{i\lambda^2}{2} \int \frac{d^4 k}{(2\pi)^4} \frac{i}{(p_i - k)^2 + i\epsilon} \frac{i}{k^2 + i\epsilon} + \dots$$

momentum space Feynman rules:

- Internal lines get propagators

$$\frac{i}{p^2 - m^2 + i\epsilon}$$

- vertices come from Lagrangian interactions & get coupling constants

times:

- lines connected to ext. points don't get propagators (connected by LSZ terms)

by LSZ terms)

- momentum conserved at each vertex

- Integrate over undetermined k - momenta.

- Sum over all diagrams

2.2.1 Signs of momenta

- No confusion the choice of which dir. momenta go

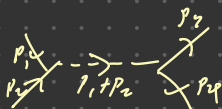
???

external momenta makes sense to assign physical values

→ δ -con. $\{ p_i = \{ p, 0 \}$ appears in δ lines
like $\delta^4(p_i - p_f)$

- Interval 1ns , integrate over p so doesn't matter sign

- Arrows drawn for neutrals p -flow



- arrows point opp. dir. for antineutrinos

2.2.2 Disconnected Graphs.



only need to really ever compute

$$\langle 0 | T \{ \mathcal{E}(\alpha_1) - \mathcal{E}(\alpha_2) \} | 0 \rangle_{\text{connected}}$$

- Everything else factored out on hermitian

2.4) Examples

$$\text{Ex: } \mathcal{L} = -\frac{1}{2} \phi \square \phi - \frac{1}{2} m^2 \phi^2 + \frac{g}{3!} \phi^3$$

cross section:

$$\frac{d\sigma}{d\Omega}(\phi\phi \rightarrow \phi\phi) = \frac{1}{64\pi^2 E_{cm}^2} (M)^2$$

incoming: p_1^μ & p_2^μ

outgoing: p_3^μ & p_4^μ

3 diagrams:

$$iM_3 = \text{Diagram 1} = (ig) \frac{i}{(p_1+p_2)^2 - m^2 + i\epsilon} (ig) = \frac{-ig^2}{s - m^2 + i\epsilon}$$

$s = (p_1 + p_2)^2$

2nd:

$$\text{Diagram 2} = (ig) \frac{i}{(p_1-p_3)^2 - m^2 + i\epsilon} = \frac{-ig^2}{t - m^2 + i\epsilon}$$

$t = (p_1 - p_3)^2$

3rd:

$$iM_3 = \text{Diagram 3} = (ig) \frac{i}{(p_1-p_4)^2 - m^2 + i\epsilon} (ig) = \frac{-ig^2}{u - m^2 + i\epsilon}$$

$u = (p_1 - p_4)^2$

$$\rightarrow \frac{d\sigma}{d\Omega}(\phi\phi \rightarrow \phi\phi) = \frac{g^4}{64\pi^2 E_{cm}^2} \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right]^2$$

(³tree level) so loop (1 & 2 loops)

Mandelstam Variables:

s, t, u from before

defined as: ($e \rightarrow e$ scattering)

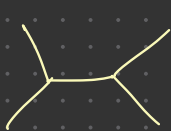
$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2$$

$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2$$

$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2$$

rel. to G_{ij} $s + t + u = \sum_j m_j^2$

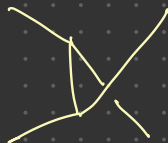
$e \rightarrow e$ scattering channels:



s-channel



t-channel



u-channel

- $s\text{-channel}: p_m^2 = s > 0$

- $t \neq u: t < 0 \neq u < 0$

- s, t, u not inv.

1.4.12) Derivative Couplings

4 ps. Limit = $\lambda \phi_1 (\partial_\mu \phi_2) (\partial_\mu \phi_3)$

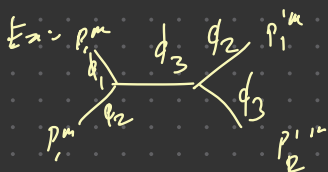
- ∂_μ give momentum factors

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \left(a_p e^{-ipx} + a_p^\dagger e^{ipx} \right)$$

→ if particle created/changes from vertex, gets $i p_\mu$ factor

if particle enters vertex gets destroyed → $-i p_\mu$ factor

- $\mathcal{L}$ must be deep track of momentum flow



exchanged p is $k^\mu = p_1^\mu + p_2^\mu = p_1'^\mu + p_2'^\mu$

$$\mathcal{M} = (i\lambda)^2 (-i p_2^\mu) (i p_1^\mu) \frac{i}{k^2} (i p_2'^\mu) (-i p_1'^\mu) = \frac{-i\lambda^2 [p_2 \cdot p_1 + p_2^2] [p_2' \cdot p_1' + p_2'^2]}{(p_1 + p_2)^2}$$

note: should get same ans. if using Feynman rules
 splits by int. by first 2 lines

$$\text{Limit} = -\lambda \phi_3 [(\partial_\mu \phi_1) (\partial_\mu \phi_2) + \phi_1 \square \phi_2]$$

which it does

Generalization:

$$L_{int} = \mathcal{O}_i(\phi_1, \dots, \phi_n)$$

- Give contribution from derivative on each field (free p)

if vertex gives ψ info derivative,

it would give

$$\left(\sum_{\text{inc.}} p_{\mu}^i - \sum_{\text{outgo.}} p_{\mu}^j \right) \psi$$

- total derivatives don't contribute to mat. elements
(in perturbation theory *int*)

$$e.g.: \text{protons} \quad \int d^3x \bar{\psi} \gamma_{\mu} \psi = 4 \partial_{\mu} (\epsilon^{\text{protons}} A_{\mu} \psi \psi)$$

total deriv:-

- has physical effects

- would be an observable dipole for neutron (electric)

- Also mass of η meson

Normal ordering & Wick's Theorem:

Normal Order:

all $a_p^\dagger$ are on the left of a_p operators

(use colours)

$$\rightarrow = (a_p^\dagger + a_p) (a_k^\dagger + a_k) = a_k^\dagger a_p + a_p^\dagger a_k + a_p a_k + a_k a_p^\dagger$$

$$\rightarrow \langle 0 | : \phi(x_1) \dots \phi(x_n) : | 0 \rangle = 0$$

only c-numbered functions ^f remain
not func f:

$$\langle 0 | f | 0 \rangle = f$$

Note:

$$T \{ \phi(x) \phi_0(y) \} = : \phi_0(x) \phi_0(y) : + P(x, y) :$$

Wick's theorem:

$$T \{ \phi_+(x_1) \dots \phi_0(x_n) \} = : \phi_0(x_1) \dots \phi_0(x_n) : + \text{all possible contractions}$$

proof:

$$\phi_0(x) = \phi_+(x) + \phi_-(x)$$

$$\phi_+(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} a_p^\dagger e^{ipx}, \quad \phi_-(x) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} a_p e^{-ipx}$$

[- since $\{a_k, a_p^\dagger\} = (2\pi)^3 \delta^3(\vec{p} - \vec{k})$, commutators of these operators just functions ???]

$$\eta_p(x_1, x_2) = \langle 0 | T \{ \phi(x_1), \phi_0(x_2) \} | 0 \rangle$$

$$= [\phi_-(x_1), \phi_+(x_2)] \theta(t_1 - t_2) +$$

$$[\phi_-(x_2), \phi_+(x_1)] \theta(t_2 - t_1)$$

$$t_1 > t_2,$$

$$\begin{aligned} T \{ \phi_0(n_1) \phi_0(n_2) \} &= \phi_+(n_1) \phi_+(n_2) + \\ &\quad \phi_+(n_1) \phi_-(n_2) \\ &\quad + \phi_-(n_1) \phi_+(n_2) \\ &\quad + \phi_-(n_1) \phi_-(n_2) \end{aligned}$$

$$\begin{aligned} \rightarrow T \{ \phi_0(n_1) \phi_0(n_2) \} &= \\ &= \phi_0(n_1) \phi_0(n_2) + \{ \phi_-(n_1), \phi_+(n_2) \} \\ &\quad A_1 > t_2 \end{aligned}$$

then

$$T \{ \phi_0(n_1) \phi_0(n_2) \} =$$

$$: \phi_0(n_1) \phi_0(n_2) : + p_+(n_1, n_2)$$

Full proof uses induction and gives

$$\begin{aligned} T \{ \phi_0(n_1) \phi_0(n_2) \dots \phi_0(n_n) \} &= \{ \phi_+(n_1) + \phi_-(n_1) \} : \phi_0(n_2) \dots \phi_0(n_n) \\ &\quad + \text{all possible contractions} : \end{aligned}$$

2.1)

$$L = -\frac{1}{2} \phi(0, \vec{x}) \phi + \frac{g}{3!} \phi^3$$